

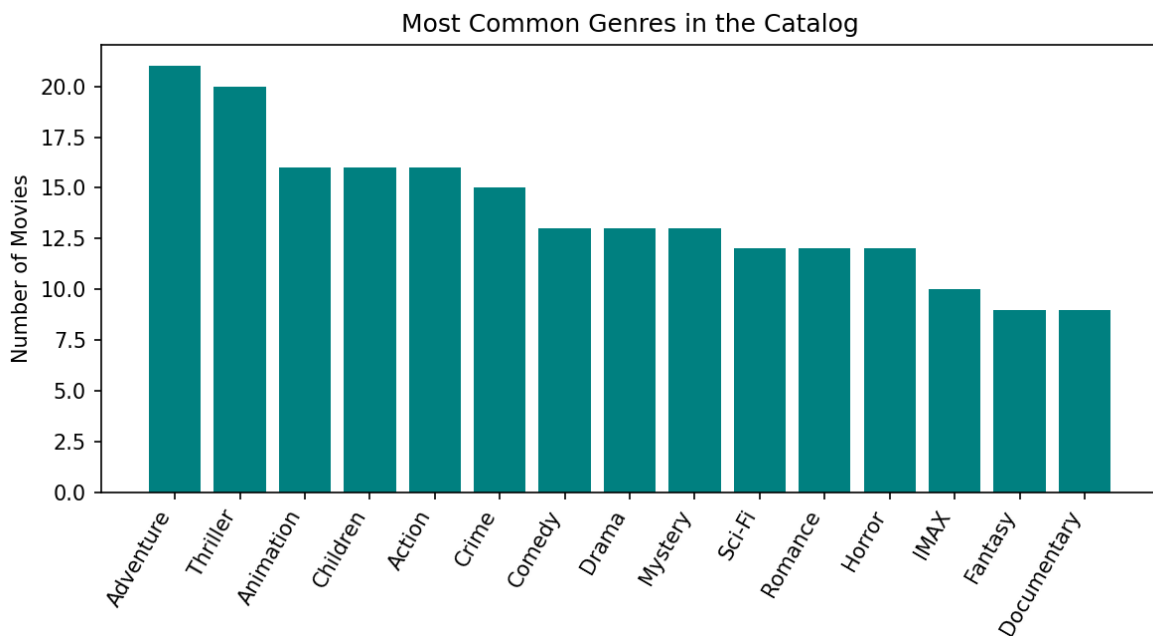
Content-Based Movie Recommendation System

Built on the MovieLens dataset (100 movies). Recommendations are generated using TF-IDF vectorization of movie genres (and tags, if available) followed by cosine similarity between movies — no user ratings are needed, so this works even for brand-new titles.

Methodology

1. Clean the movie catalog and combine each movie's genres (and tags) into one text field.
2. Vectorize that text with TF-IDF (term frequency-inverse document frequency), turning each movie into a numeric vector where rarer, more distinctive genres/tags carry more weight.
3. Compute pairwise Cosine Similarity across all movie vectors — a score from 0 (nothing in common) to 1 (identical content profile).
4. For a given movie, rank every other movie by similarity score and return the top N.

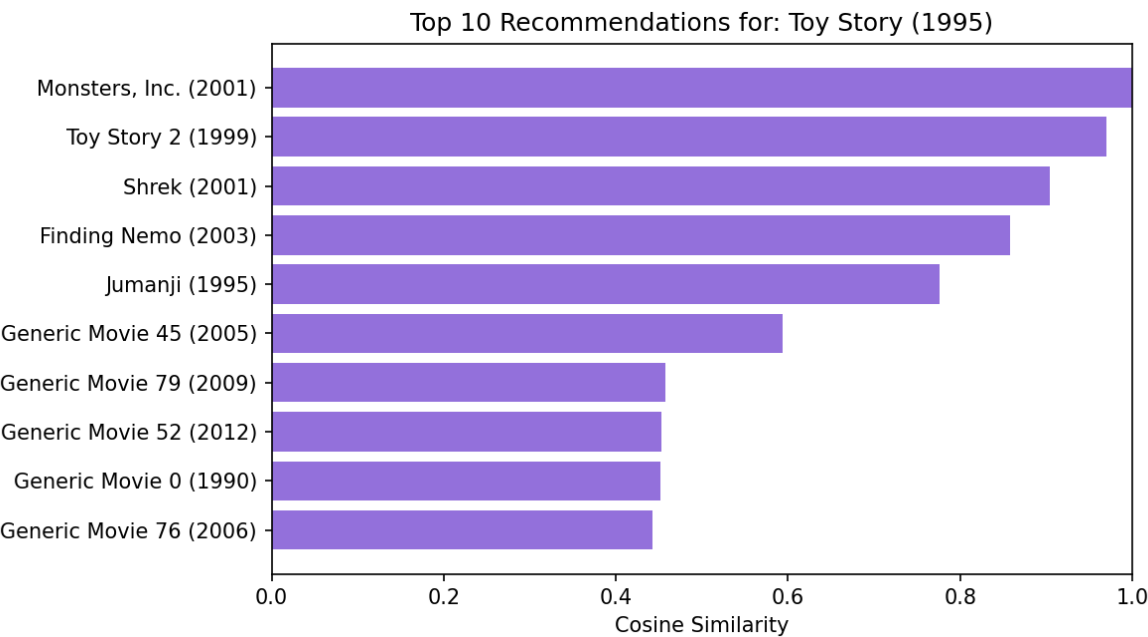
Catalog Overview



Example Recommendations

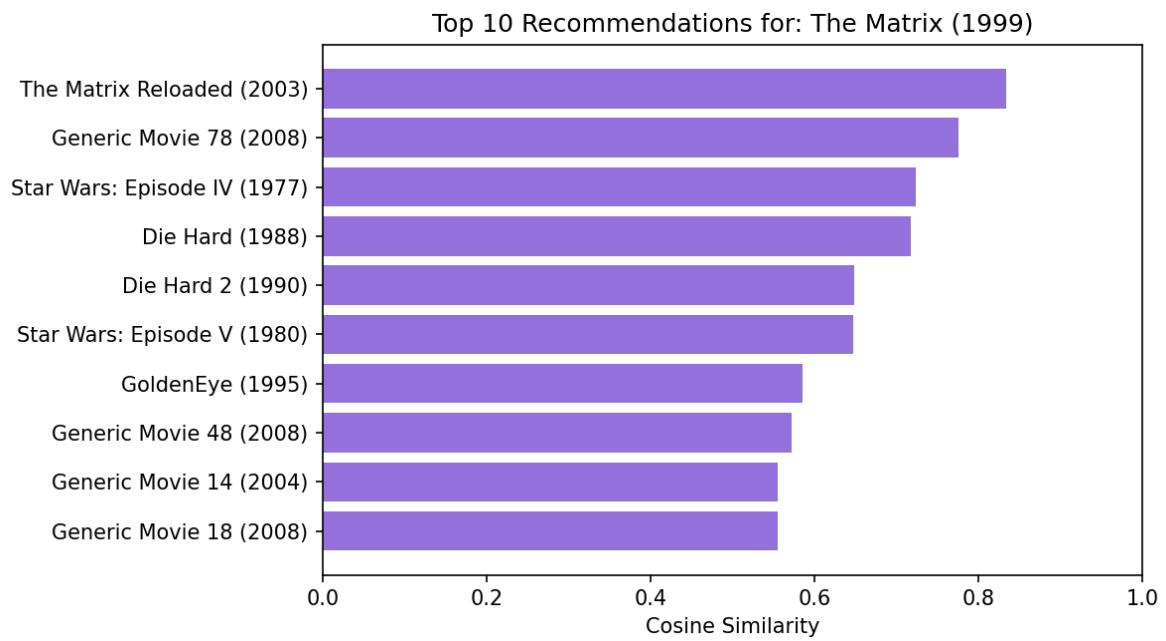
The following examples show recommendations generated for a few sample movies from the catalog, to illustrate how the system behaves.

Because you watched: “Toy Story (1995)”



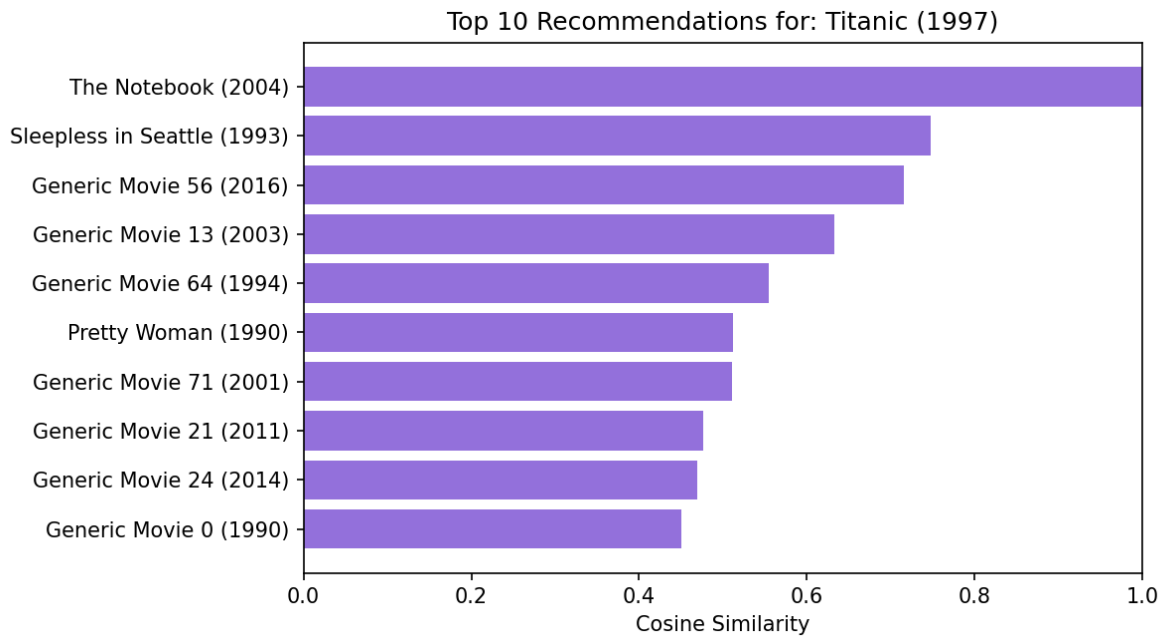
#	Recommended Movie	Genres	Similarity
1	Monsters, Inc. (2001)	Adventure Animation Children Comedy Fantasy	1.000
2	Toy Story 2 (1999)	Adventure Animation Children Comedy Fantasy	0.970
3	Shrek (2001)	Adventure Animation Children Comedy Fantasy Romance	0.904
4	Finding Nemo (2003)	Adventure Animation Children Comedy	0.858
5	Jumanji (1995)	Adventure Children Fantasy	0.776

Because you watched: “The Matrix (1999)”



#	Recommended Movie	Genres	Similarity
1	The Matrix Reloaded (2003)	Action Sci-Fi Thriller IMAX	0.834
2	Generic Movie 78 (2008)	Sci-Fi Action	0.776
3	Star Wars: Episode IV (1977)	Action Adventure Sci-Fi	0.724
4	Die Hard (1988)	Action Thriller	0.717
5	Die Hard 2 (1990)	Action Adventure Thriller	0.648

Because you watched: “Titanic (1997)”



#	Recommended Movie	Genres	Similarity
1	The Notebook (2004)	Drama Romance	1.000
2	Sleepless in Seattle (1993)	Comedy Drama Romance	0.748
3	Generic Movie 56 (2016)	Romance	0.716
4	Generic Movie 13 (2003)	Romance	0.633
5	Generic Movie 64 (1994)	Drama	0.555

Note: This is a content-based recommender using genres/tags only — it has no knowledge of user ratings or viewing behavior, so it will always recommend movies with similar content regardless of popularity. Combining this with a collaborative-filtering model (based on user ratings) is the natural next step.